

POCKET SMART AI — Your Smart Budget & Recommendation Assistant

Phase 2: Requirement Analysis

St. George's Arts and Science College (UNM1727) · TN Skills | Generative AI · Team ID: SWTID-2026-2087

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2.1 Purpose of This Phase

This phase converts the ideation-stage concept into a concrete, testable set of requirements: the problem statement, functional requirements, non-functional requirements, prerequisites, and the prior knowledge needed before development begins.

2.2 Problem Definition

Users planning multiple purchases often need to make several connected decisions under one fixed budget. A useful system should support planning before spending, preserve the total budget constraint, and present the result in a way that is easy to understand — even for a non-technical user.

2.3 Functional Requirements

ID	Requirement
FR-01	Provide an application entry point (landing page).
FR-02	Support a user access/registration flow where implemented.
FR-03	Accept budget and requirement inputs from the user.
FR-04	Validate input values before processing.
FR-05	Process the budget allocation across the selected categories.
FR-06	Generate recommendations or purchasing guidance.
FR-07	Present results through the application interface.
FR-08	Handle invalid or incomplete input without crashing.

2.4 Non-Functional Requirements

- **Usability** — the input forms and results must be understandable to a non-technical user.
- **Responsiveness** — the interface should react to input without unnecessary delay.
- **Maintainability** — logic should be organized into clear, separable components.
- **Reliability** — the system should behave predictably across repeated use.
- **Basic security hygiene** — credentials and secrets must not be exposed.
- **Predictable validation** — the same invalid input should always be rejected the same way.
- **Readable output** — allocation and recommendation results should be presented clearly.
- **Deployment reproducibility** — the project should be able to be rebuilt and redeployed reliably.

2.5 Prerequisites Identified

Software Requirements

A modern web browser; a code editor/IDE; the runtime required by the selected implementation; a package/dependency manager; version-control tooling; and a hosting account for online deployment.

Libraries / Frameworks / Dependencies

The exact dependency list is taken from the project's actual configuration file (requirements.txt) rather than assumed generic package names.

Hardware Requirements

A standard laptop or desktop capable of running a browser and the development runtime, with a stable internet connection recommended for hosted services. No special sensors or dedicated hardware are required.

2.6 Prior Knowledge Required

Area	Concepts Needed
Programming	Basic logic, variables, conditions, functions, data structures, debugging.
Web Development	HTML/CSS/JavaScript or equivalent frontend technology; form handling; browser concepts.
Backend Concepts	Request/response flow, routes/controllers, validation, session/authentication, server-side processing.
Database Fundamentals	Tables/collections, records, identifiers, CRUD operations, data validation.
Networking / Cloud	HTTP/HTTPS, localhost, ports, domains, hosting, deployment concepts.
AI/ML Fundamentals	Basic understanding of recommendation logic, prompts/models, or rule-based decision support.

2.7 Target Users & Stakeholders

- Students, individual consumers, and households planning purchases.
- Event planners distributing a fixed amount across event categories.
- Home-decor planners allocating budget across furnishing categories.
- The academic evaluators of the TN Skills | Generative AI project submission.

2.8 Outcome of This Phase

A documented set of functional requirements (FR-01 to FR-08), non-functional quality expectations, prerequisites, and prior-knowledge checklist, forming the baseline against which the design and later testing phases are validated.